TOOL WIREFRAME

Bayer Crop Science: Yield Simulator

Navigation diagram: primary flow

The primary flow represents the journey of the primary user (João) through the simulator from first arrival to completed output. Each node is shown with its identifier and canonical name. Transitions are annotated with the user action or system event that triggers them.

[P1] **Home**
Purpose statement, scope declaration, usage sequence, and 'Start Configuration' call to action.

↓ *User reads purpose statement and clicks 'Start Configuration'* ↓

[P2] **Variable Configuration**
Pre-filled inputs from FieldView dataset. Manual climate inputs. Soil and equipment panels. 'Run Simulation' button.

↓ *User confirms all inputs and clicks 'Run Simulation' — all three required fields are filled* ↓

[S1] **Simulation Processing**
Progress indicator with step labels: 'Loading historical data', 'Running model', 'Preparing comparable cases'. Duration: 10 to 60 seconds.

↓ *Model computation completes successfully* ↓

[P3] **Behavior Demonstration**
Three scenario panels. CV and Phenotypic Plasticity metrics. Comparable cases section. Narrative recommendation. Model Assumptions panel.

Navigation diagram: alternative routes

The following diagram shows all alternative routes, error states, and recovery paths that the navigation must support. Each alternative route is triggered by a specific condition and leads to a defined resolution.

From	Condition	System state	What the user sees	Resolution path
P2	User clicks 'Run Simulation' with planting date, rainfall, or	**S2: Incomplete Configuration**	The three required fields are highlighted in red with inline error messages. A summary banner at the top of the	User fills the missing fields. As soon as all three are present, the banner clears and

From	Condition	System state	What the user sees	Resolution path
	temperature missing		form lists the missing fields by name.	'Run Simulation' becomes active.
P2	User selects a state outside MG, PR, MS, GO	**S4: Field Out of Scope**	The state selector shows a tooltip: 'This state is outside the current scope (MG, PR, MS, GO). The simulator cannot run for this location.' The 'Run Simulation' button remains disabled.	User selects an in-scope state. The form re-fills with historical data for the selected state.
P3	Similarity search returns fewer than 3 comparable cases	**S3: No Comparable Cases**	The comparable cases section displays: 'No sufficiently similar historical records were found for this configuration. Adjust planting date, soil profile, or seed population to broaden the search.' The rest of P3 loads normally.	User clicks 'Edit configuration' link, returns to P2, adjusts one or more inputs, and re-runs the simulation.
P3	User clicks 'Edit configuration'	**P2 (preserved state)**	P2 loads with all previously entered values intact. A banner at the top reads: 'You are editing a previous configuration. Your changes will update the simulation on Page 3.'	User modifies one or more values and clicks 'Run Simulation' again. S1 runs and P3 updates.

Transition logic: what each action does

This section documents some possible user action that causes a navigation transition, the preconditions that must be true for the action to be available, and the system behaviour that follows.

Action label	Located on	Precondition	System behaviour
'Start Configuration' button	P1: Home	None; always active	Navigates to P2. If a previous session exists with saved values, P2 loads with those values pre-filled and a banner noting that previous values were restored.

Action label	Located on	Precondition	System behaviour
'Run Simulation' button	P2: Variable Configuration	All three required fields filled: state, crop, and planting_date.	Triggers S1 (Simulation Processing). P2 values are frozen and passed to the model. Progress indicator displays three sequential steps.
'Run Simulation' button (blocked)	P2: Variable Configuration	One or more required fields are empty, or state is out of scope.	Button is visually disabled (grayed out, not clickable). Missing fields are highlighted. An inline message near the button reads: 'Fill in the required fields above to run the simulation.'
'Edit configuration' link	P3: Behavior Demonstration	None; always active on P3.	Navigates back to P2. All P2 values are restored exactly as they were when the simulation was last run. A banner at top of P2 notifies the user they are editing a previous configuration.
'Return to Home' link	P3: Behavior Demonstration	None; always active on P3.	Navigates to P1. Session values in P2 are preserved but not highlighted. The user may resume via 'Start Configuration'.
State selector: out-of-scope option selected	P2: Variable Configuration	User selects a state other than MG, PR, MS, GO.	Triggers S4 inline. A tooltip appears on the selector. The 'Run Simulation' button is disabled. No data is loaded for the out-of-scope state.
Comparable case: 'View details' link	P3: Behavior Demonstration	Comparable case is present in the results panel.	Expands the case card in place to show all matching variable values and their observed outcome. Does not navigate away from P3.

Screen content overview

Each primary screen (P1, P2, P3) is described below with its content sections, the purpose of each section, the user action it enables, and the design principles from the persona document that govern its layout. This overview provides the direct specification that the wireframe screens must implement.

P1: Home

Section	Content
Simulator identity	Name of the simulator, Bayer branding, sprint and group identifier.
Purpose statement	Two to three sentences explaining what the simulator does and what question it answers: 'This tool helps crop managers estimate the yield range for a planned field season by combining your FieldView planting history with agronomic model rules.'
Scope declaration	Explicit statement of the four supported states (MG, PR, MS, GO) and two supported crops (corn and soybeans). One sentence explaining that climate inputs must be entered manually.
Usage sequence	A three-step sequence (numbered or visually stepped): Step 1: review and configure your field variables on the next page. Step 2: run the simulation. Step 3: review your scenarios, comparable cases, and recommendation.
'Start Configuration' call to action	A single prominent button navigating to P2.

P2: Variable Configuration

Section	Content
Field identity header	Field name (read-only), field area, planted area. FieldView attribution label.
Scope selectors	State dropdown (MG, PR, MS, GO only), crop dropdown (corn or soybeans), season code toggle (Summer, Safrinha, Winter), crop year input.
Planting inputs	Planting date (primary editable, required), seed product selector, seed population per hectare (numeric, pre-filled, editable).
Soil profile panel	Soil order, soil texture, soil pH range (min and max), drainage class. All pre-filled from FieldView; all overridable.
Equipment context panel	Planter manufacturer, planter model. Read-only. Small visual weight.
'Run Simulation' action area	Primary button 'Run Simulation'. Disabled state with inline message when required fields are missing.

P3: Behavior Demonstration

Section	Content
Configuration summary banner	Read-only summary of the key inputs used: state, crop, planting date, season code. 'Edit configuration' link.
Three scenario panels	Pessimistic, Most Likely, and Optimistic yield estimates in sc/ha. Each panel shows the yield value, a confidence band, and a one-line summary of the dominant factor driving that scenario.
Simulation metrics	CV of Seed Distribution (percentage with threshold annotation), Phenotypic Plasticity index (tier label plus numeric index). Reliability flags when fallback data was used.
Narrative recommendation	3 to 5 sentence text block. Written in second person. Specific to the inputs used and the scenarios computed.
Dataset similarity search section	Up to 5 comparable historical cases from the FieldView dataset. Each case card shows: state, crop, planting month, soil order, drainage class, population band, and observed outcome. Matching criteria displayed on each card.
Model Assumptions panel	Two sub-lists: Immutable premises (lock icon) listing Embrapa thresholds and scenario rules. Auditable metrics (edit icon) listing all Page 2 inputs with their confirmed values.
Navigation footer	'Edit configuration' link (returns to P2 with preserved values) and 'Return to Home' link.

Full variable-to-interface traceability table

Each row below represents one variable that appears in the simulator interface. Columns describe its origin, the page and element where it appears, how it is populated, its update frequency, and the fallback behaviour when the value is absent.

Variable shown to user	Interface element	Fallback if absent
Page 2: Scope selectors		
State	Dropdown selector; only MG, PR, MS, GO enabled	Selector defaults to MG; out-of-scope states grayed out with tooltip explaining constraint
Crop	Dropdown selector; corn and soybeans only	Defaults to soybeans; no fallback needed since both values are always present
Season code	Three-option toggle: Summer / Safrinha / Winter	Defaults to Summer (SUM) if no prior record; displays note that value is estimated

Variable shown to user	Interface element	Fallback if absent
Page 2: Field identity		
Field name	Read-only label at top of form	Displays 'Unnamed field' with field_uuid in small text
Field area (ha)	Read-only numeric display, one decimal	Displays 'Area not recorded'; planted_area_ha used as proxy if available
Planted area (ha)	Read-only numeric display, one decimal	Displays dash; simulation proceeds without this value
Page 2: Planting inputs		
Planting date	Date picker; pre-filled, fully editable	Displays empty date picker with placeholder 'Enter planting date'; field is required; simulation blocked until filled
Seed population (seeds/ha)	Numeric input with unit label; pre-filled, editable	Displays median for state and crop as default value; note: 'Using regional median — confirm or adjust'
Page 2: Soil profile		
Soil order	Dropdown with SiBCS classification options	Displays 'Not identified'; simulation uses regional modal class; note displayed
Soil texture	Dropdown: argilosa, franco-argilosa, arenosa, etc.	Displays 'Not identified'; treated as clay loam in model; note displayed
Soil drainage class	Dropdown: bem drenado, moderado, imperfeitamente drenado	Defaults to 'bem drenado'; note displayed
Page 3: Simulation outputs		
Pessimistic scenario yield	Large numeric display, sc/ha; red accent	Not applicable; always computed
Most likely scenario yield	Large numeric display, sc/ha; neutral accent	Not applicable; always computed
Optimistic scenario yield	Large numeric display, sc/ha; green accent	Not applicable; always computed
CV of Seed Distribution (%)	Numeric metric with threshold annotation	If seed population is absent, metric is flagged as 'Estimated from regional median: lower reliability'

Variable shown to user	Interface element	Fallback if absent
Phenotypic Plasticity index	Tier label (Low, Moderate, High) plus numeric index	If soil data is partial, tier is displayed with reliability flag
Narrative recommendation	Text block, 3 to 5 sentences, prominent typography	Not applicable; always generated
Page 3: Behavior Demonstration — Dataset similarity search		
Comparable case: state and crop	Label in each comparable-case card	If fewer than 3 matches found, section displays 'Insufficient comparable cases in dataset for this configuration'
Comparable case: planting window	Label in each comparable-case card	Same as above
Comparable case: soil match	Label in each comparable-case card	Same as above
Comparable case: population match	Label in each comparable-case card	Same as above
Comparable case: seed product	Optional label; shown only when match is exact	Not shown if no exact match; does not affect case display
Page 3: Behavior Demonstration — Model assumptions panel		
Immutable premises	Locked-icon list in 'Model Assumptions' panel	Not applicable; always displayed
Auditable metrics	Edit-icon list linking back to Page 2	Not applicable; always displayed